Design a seven-segment code

Task 4.1 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Lay out a figure 8 with A across the top, B upper right, C lower right, D along the bottom, E lower left, F upper left and G across the middle. Agree on 1 = lit and 0 = dark. Draw the switched lamp loop.

Mark the segment positions A–G in the same order on two cards. Count which segments light for one digit.

Digit	A B C D E F G	Digit	A B C D E F G
0		5	
1		6	
2		7	
3		8	
4		9	

02 Plan it on paper

Make an on/off code for digits 0–9. Agree on the bit order and the meaning of 1 and 0.

My prediction, chosen feature or design limit

Design a seven-segment code

Task 4.1 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Build a switched lamp circuit. One partner sends three secret digit patterns and the other reconstructs the display. Swap roles and decode three new patterns. If a digit is wrong, compare segment order before changing the key.

Round	Sent digit / bits	Decoded digit	Match or correction
A1			
A2			
A3			
B1			
B2			
B3			

04 Use the reading

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Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Show the code for one digit and explain how another person can decode it without seeing your drawing.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.